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ABSTRACT

This report presents the work completed during a twelve-week Microsoft Elevate Internship (January–April 2026), undertaken as part of the GTU 8th Semester curriculum at Government Engineering College, Bharuch. The internship, conducted in collaboration with Edunet Foundation, FICE Education, and Microsoft, focused on Data Analytics, Machine Learning, Deep Learning, and Microsoft Azure Cloud technologies.

*The capstone project, titled “**Border Defence and Surveillance**” aims to enhance border security through intelligent video analytics. The system automates surveillance by processing video streams using OpenCV and performing object detection with a custom-trained YOLOv8 model across multiple classes such as person, vehicle, aircraft, and ship. Anomaly detection is implemented using Isolation Forest, followed by classification techniques to identify suspicious activities.*

The system integrates zone-based intrusion detection using geometric algorithms and performs temporal pattern analysis using multi-frame tracking. A real-time alert mechanism prioritizes threats and stores data using Azure Blob Storage and Cosmos DB. Additionally, a Streamlit-based dashboard provides visualization and monitoring capabilities for operational use.

The model was trained on a balanced dataset combining DOTA, VisDrone, and xView datasets, achieving a significant improvement in detection performance with enhanced precision and recall. The system was validated through automated testing and CI/CD integration, ensuring reliability and scalability.

This project demonstrates the practical application of Artificial Intelligence and cloud technologies in solving real-world surveillance challenges, aligning with the objectives of the GTU internship program.

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ABBREVIATIONS

Abbreviation	Full Form / Meaning
BDS	Border Defence and Surveillance
AI	Artificial Intelligence
ANN	Artificial Neural Network
API	Application Programming Interface
AZ-900	Microsoft Azure Fundamentals Certification
CI/CD	Continuous Integration / Continuous Deployment
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DAX	Data Analysis Expressions (Power BI formula language)
DL	Deep Learning
EDA	Exploratory Data Analysis
FPS	Frames Per Second
GTU	Gujarat Technological University
CE	Computer Engineering
IF	Isolation Forest (unsupervised anomaly detection algorithm)
IoU	Intersection over Union (object tracking and detection metric)
JSON	JavaScript Object Notation
mAP50	Mean Average Precision at IoU threshold 0.50
ML	Machine Learning
NMS	Non-Maximum Suppression
RF	Random Forest Classifier
GEC	Government Engineering College Bharuch
SRS	Software Requirements Specification
YOLO	You Only Look Once (real-time object detection framework)

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